

Blood Groups



Blood Groups

- The membranes of human RBCs contain a variety of antigens called **Agglutinogens** (glycoprotein in nature).
- Blood groups are classified according to the presence or absence of the agglutinogens on the RBCs.
- There are hundreds of different agglutinogens on the RBCs, but there are two main systems of blood grouping, the **ABO** system and the **Rh** system.

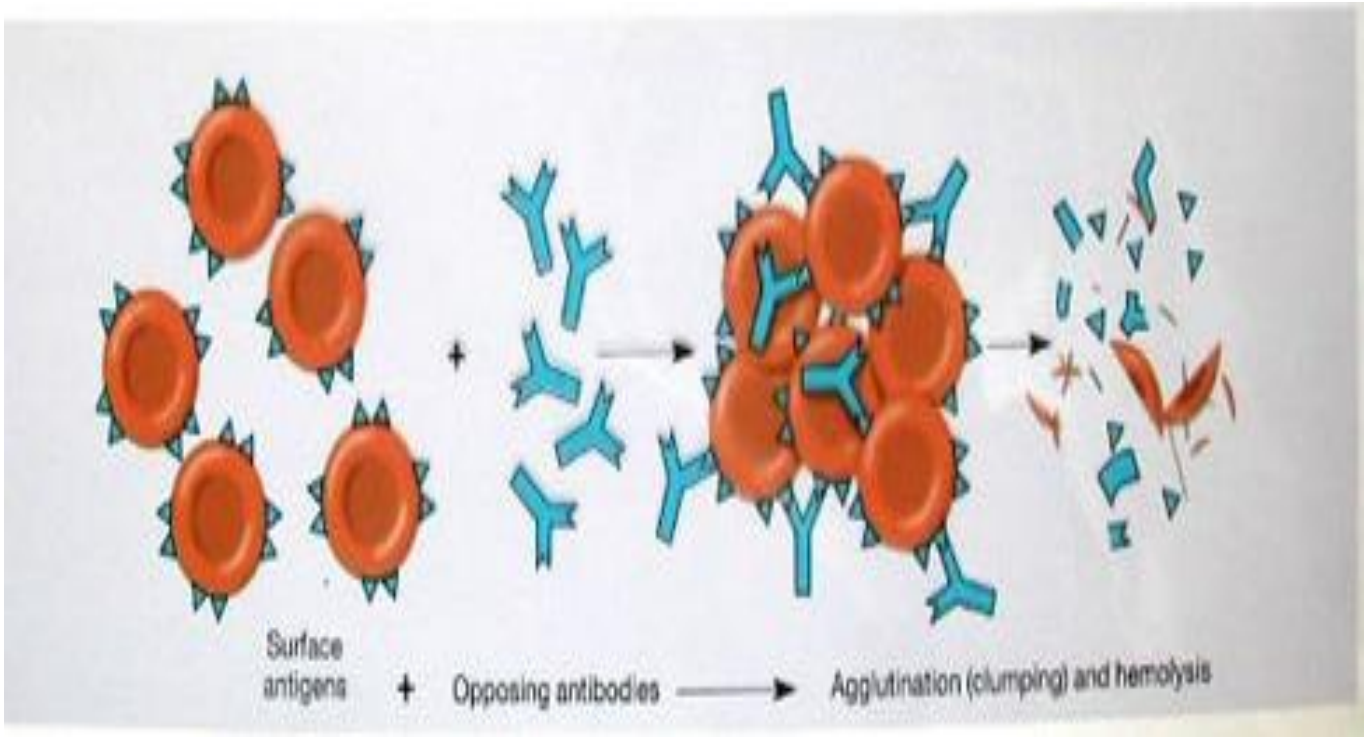


Blood Groups

- In the plasma there are antibodies called **Agglutinins**.
- As a general rule, if the RBCs of a person have any agglutinogen on their surface, his plasma **would not contain** the specific agglutinin for this agglutinogen.
- When these antigens come in contact with their specific antibodies, an antigen-antibody reaction occurs, the cells become sticky and clump together like bunches of grapes **Agglutination**.



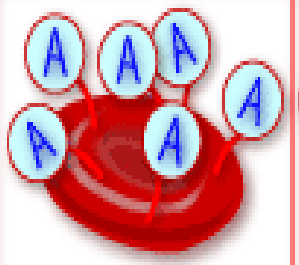
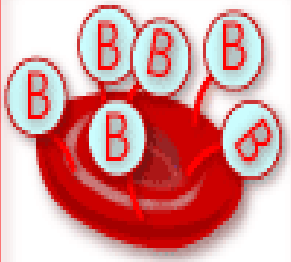
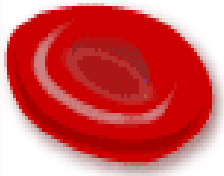
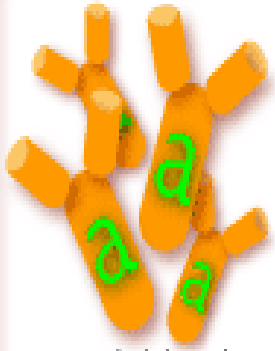
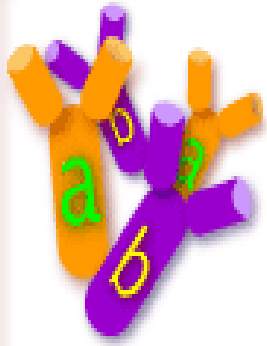
- A single antibody can attach to different red cells at the same time.
- Then destruction of the red blood cells will occur (**hemolysis**)





ABO System

The ABO Blood System

Blood Type (genotype)	Type A (AA, AO)	Type B (BB, BO)	Type AB (AB)	Type O (OO)
Red Blood Cell Surface Proteins (phenotype)	 A agglutinogens only	 B agglutinogens only	 A and B agglutinogens	 No agglutinogens
Plasma Antibodies (phenotype)	 b agglutinin only	 a agglutinin only	 No agglutinin	 a and b agglutinin



- The agglutinins are gamma globulins, **IgM** immunoglobulin molecules.

- But why are these agglutinins produced in people who do not have the respective agglutinogens in their red blood cells?

- *Small amounts of type A and B antigens enter the body in **food**, in **bacteria**, and these substances initiate the development of the anti-A and anti-B agglutinins.*



Rh system

- Rh type is the system of next importance.
- There are 6 common types of Rh antigens C, D, E and c, d, and e But, The most important and common antigen is **D.**

- Rh refers to the presence or absence of the D antigen on the red blood cell
- A person has D antigen is Rh +ve while that does not have D antigen is Rh -ve.

  85% are Rh +ve.

  15% are Rh -ve.



- Unlike the ABO blood group system, individuals who lack the D antigen do not naturally make antibodies against it.
- Production of antibody to D (**IgG**) requires exposure to the antigen.
- For this reason all individuals should be typed for D , if negative must receive Rh (D) negative blood



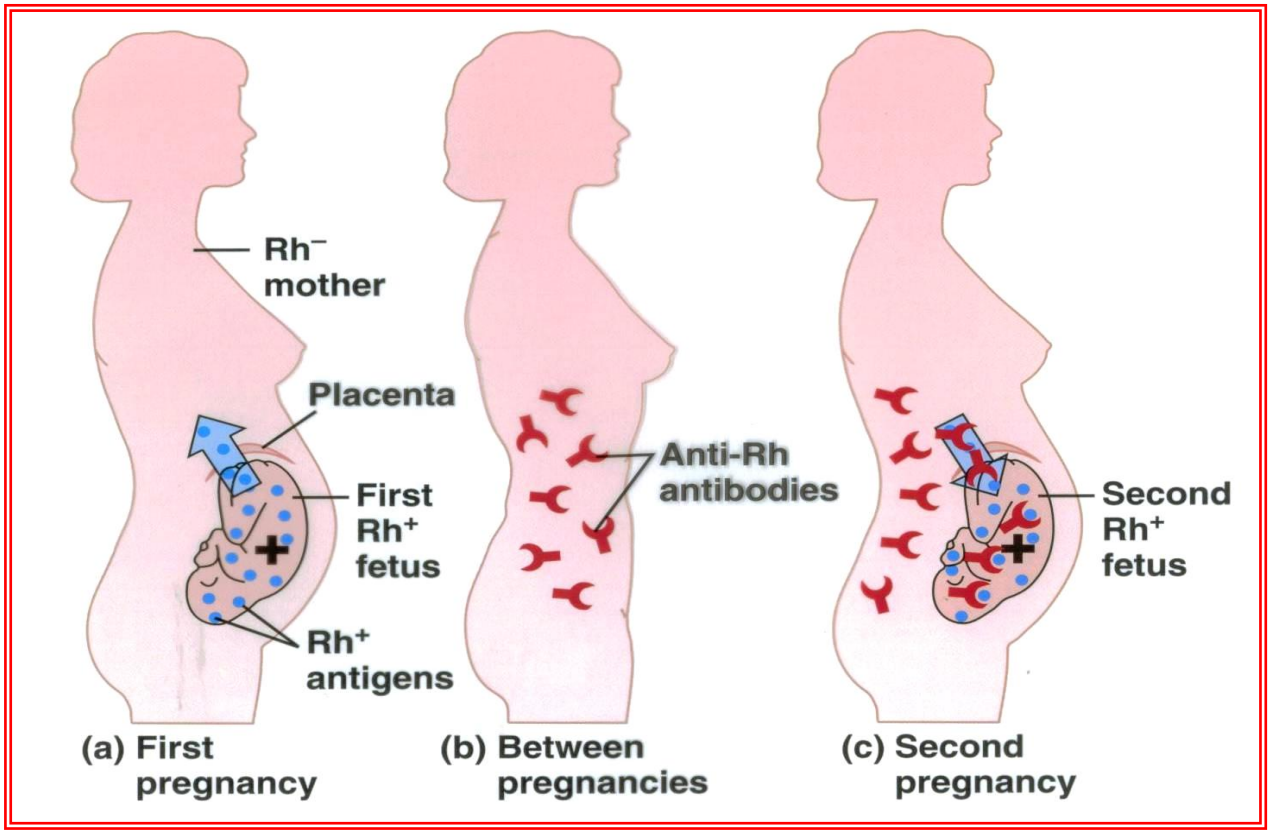
Erythroblastosis Fetalis (Hemolytic disease of the newborn)

- If the mother is Rh^- married Rh^+ man, and the baby is Rh^+. A small quantity of fetal blood leaks across the placenta into the maternal blood stream during delivery.

- The mother's immune system begins to produce anti-Rh antibodies IgG. These antibodies can cross the placenta during the subsequent pregnancy into the fetal blood.

- The 1st baby has no harm. If the 2nd fetus is Rh^+, the antigen-antibody reaction causes hemolysis of fetal RBCs and it results in HDN.




Rh⁻ mother
Placenta
First Rh⁺ fetus
Rh⁺ antigens
(a) First pregnancy
Anti-Rh antibodies
(b) Between pregnancies
Second Rh⁺ fetus
(c) Second pregnancy



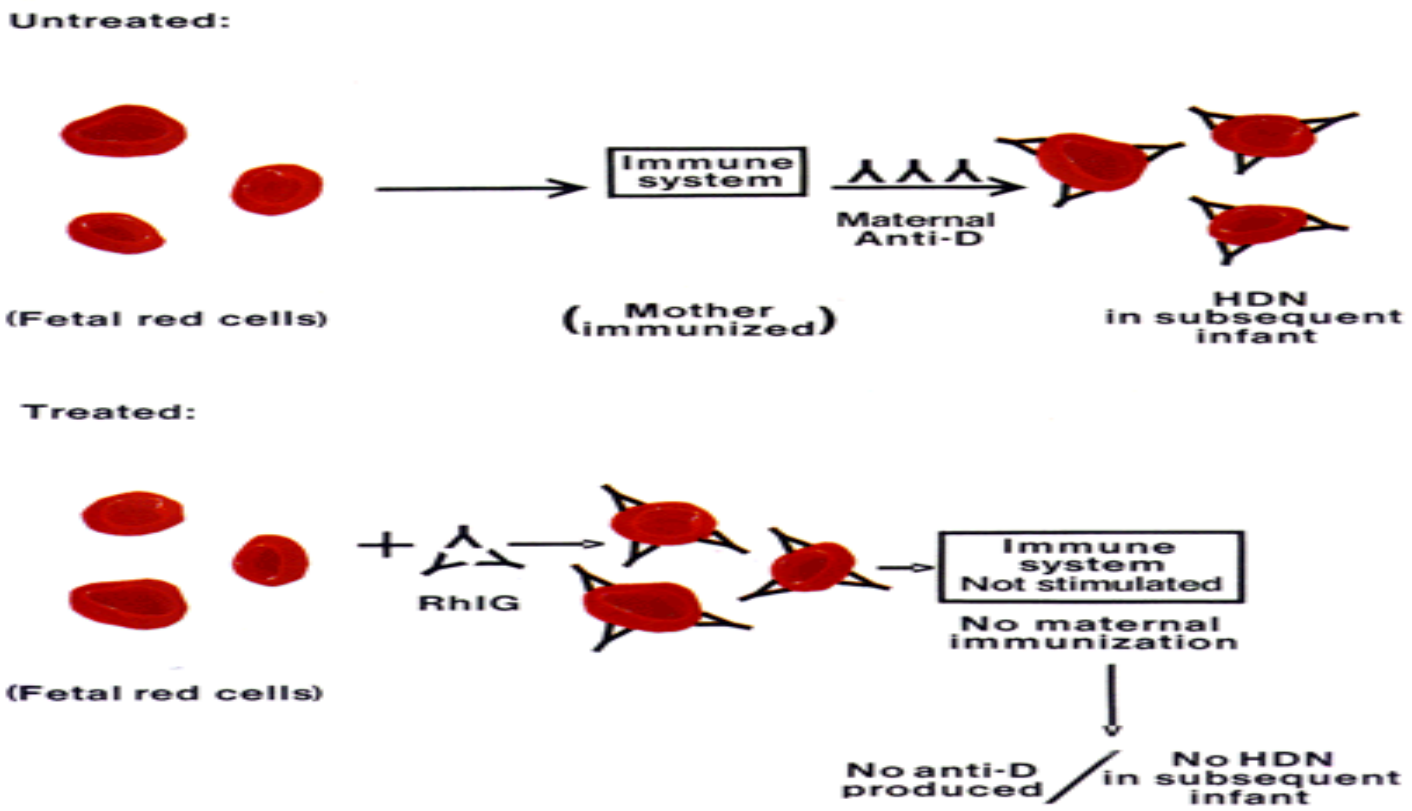
- **Clinical Features:**
  - Edema (Hydrops)
  - Jaundice
  - Anemia
  - Enlarged Liver & Spleen
  - Deposition of bilirubin in the neuronal cells leads to its destruction (Kernicterous).
  - With sever cases intrauterine fatal death occur.

- **Treatment:**
  - Exchange blood transfusion with O^- blood.

- **Prevention:**
  - Rh immune globulin (RhIG) injections to the Rh negative mother after delivery.



- Anti D binds and destroys fetal Rh D positive erythrocytes that have passed through the placenta from the fetus to the maternal circulation.
- This prevents maternal B-cell activation and memory cell formation.





Indications of blood grouping:

1. Blood transfusion.
2. Medicolegal *(good -ve &bad +ve).*
3. Pregnant ladies (erythroblastosis fetalis).



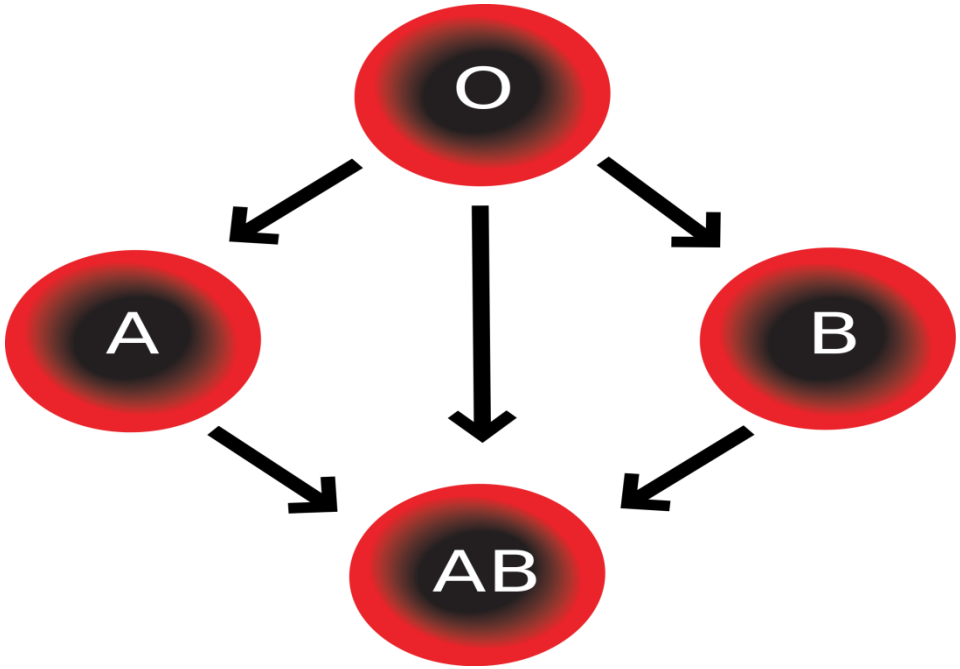
Blood Transfusion

Blood Group	Antigens	Antibodies	Can give blood to	Can receive blood from
A	A	Anti B	A and AB	A and O
B	B	Anti A	B and AB	B and O
AB	A and B	NO	AB	A,B,O and AB
O	NO	Anti A and Anti B	A,B,O, and AB	O



- Blood group O is called **"universal donor"** because it has no antigens on RBC.

- Blood group AB are called "**universal receivers**" because it has no anti-bodies in the plasma.





Indication of blood transfusion:

1. Restore blood volume as in sever hemorrhage.
2. Sever anemia.
3. Bleeding disorders e.g. purpura and hemophilia.

Types:

1. Heterologus.
2. Autologous



Precautions before blood transfusion:

1. A healthy donor free of diseases.
2. Good storage of blood with addition of Citrate, glucose, and kept at temp. of 4° C, and not to be used after 5 weeks.
3. Blood typing: they should be compatible regarding the ABO system and the Rh.

4. **Cross matching**:

Donor's RBCs + Recipient's plasma **(major cross matching)**

Recipients RBCs + Donor's plasma (minor cross matching)



Complications of blood transfusion:

Immediate

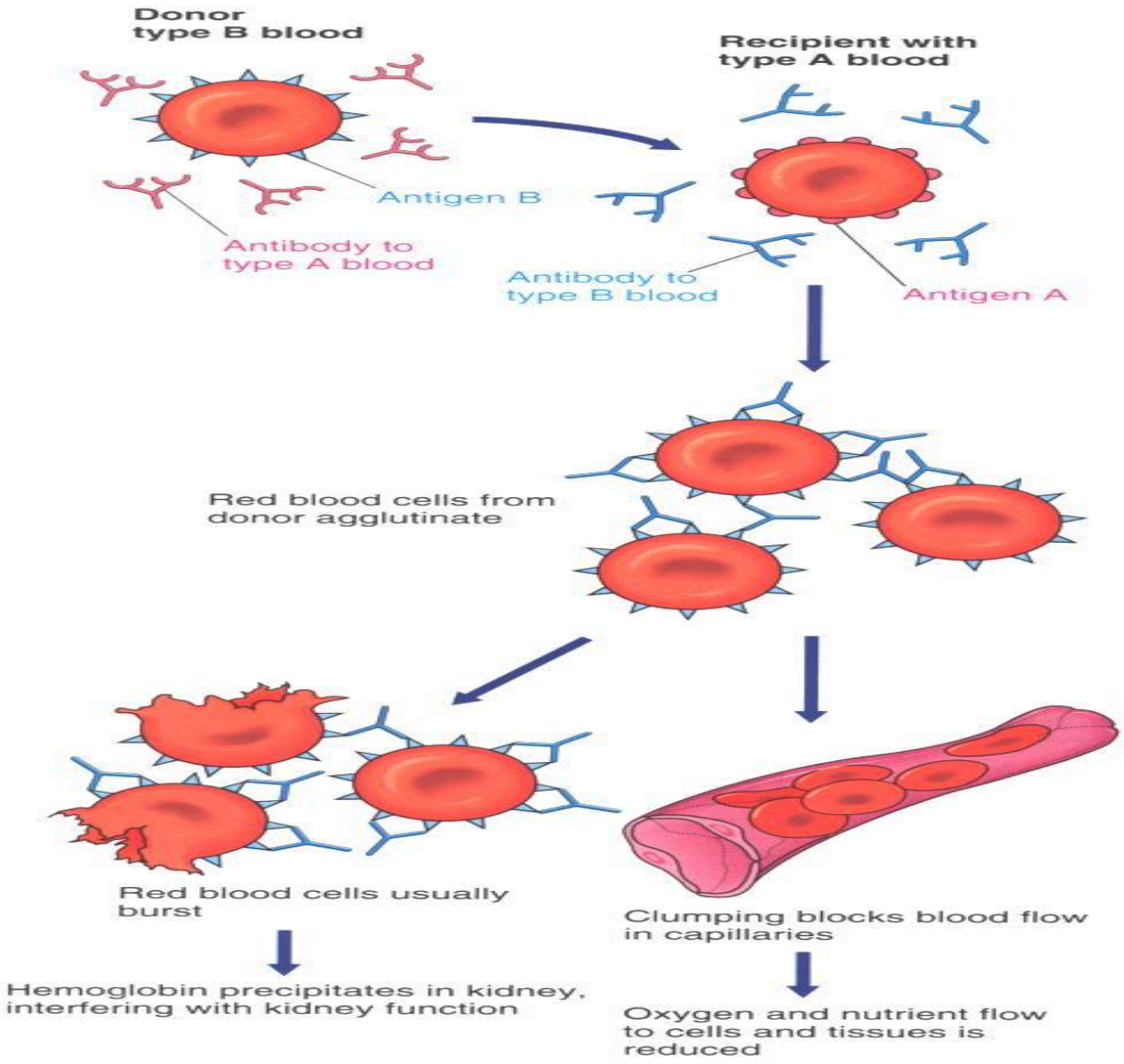
- Incompatibility
- Mechanical overload
- Hyperkalemia
- Citrate intoxication
- Bacterial contamination

Delayed

- Transmission of diseases



Incompatible blood transfusion





1. Blockage of blood capillaries caused by agglutination RBCs leading to joint pain and tightness of the chest. Followed by

2. Haemolysis which lead to
   A. Shock due to release of histamine.
   B. Hyperkalemia.
   C. Jaundice.
   D. Acute kidney shut down, which is due to:
      - Circulatory shock.
      - Toxic substance fro the Ag-Ab reaction causing powerful vasoconstriction.
      - Renal tubular blockage due to precipitation of Hb by the acidic urine in the form of acid hematin.



Blood typing by Slide method



Procedure

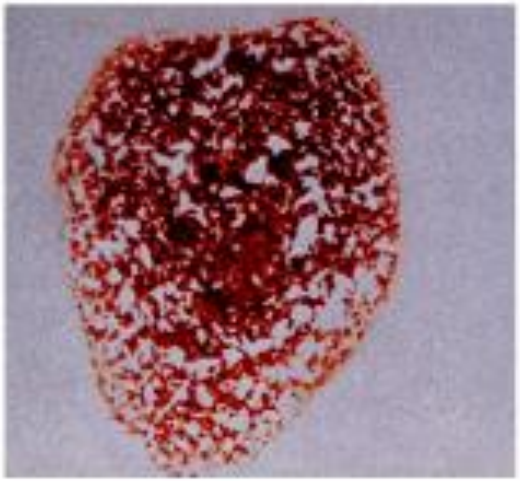
- Place one drop of anti-A, anti-B and anti-D sera on a slide.
- Puncture your thumb with a sterile lancet and allow one drop of blood to contact each of the three anti-sera.
- Avoid touching the anti-sera by your finger.
- Mix the blood with anti-sera using different small wooden sticks.
- Observe the results within 2-3 minutes

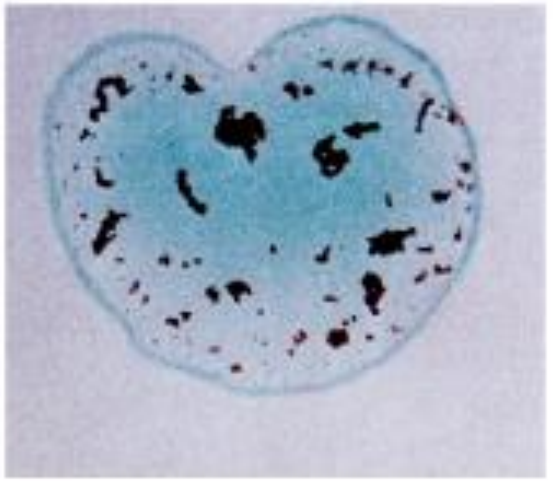

10 ml
Mediclone
A
10 ml
Mediclone
B
10 ml
Mediclone
D



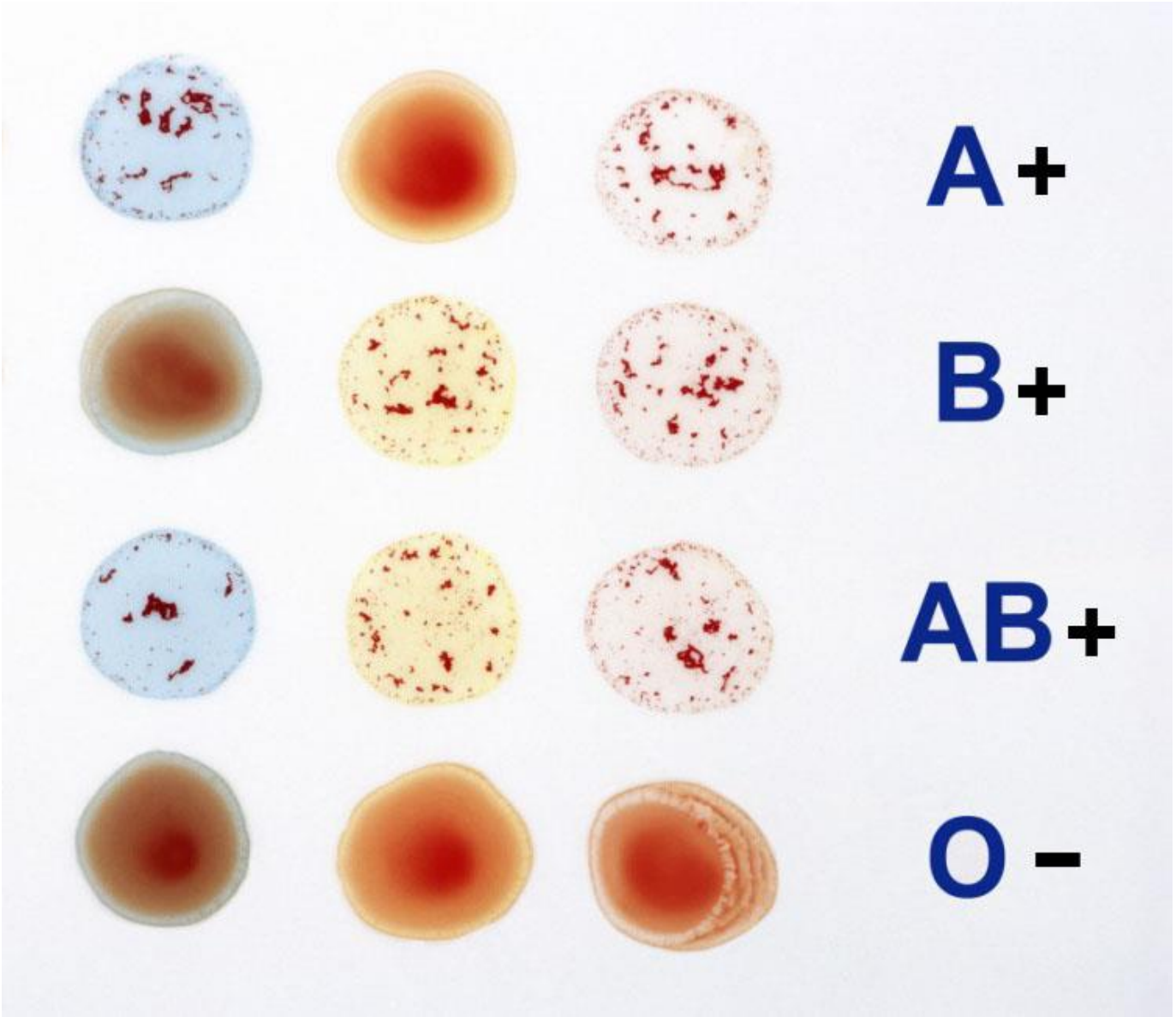
Observation:

- Agglutination of the RBCs is indicated by grainy (granular) appearance of the blood sample



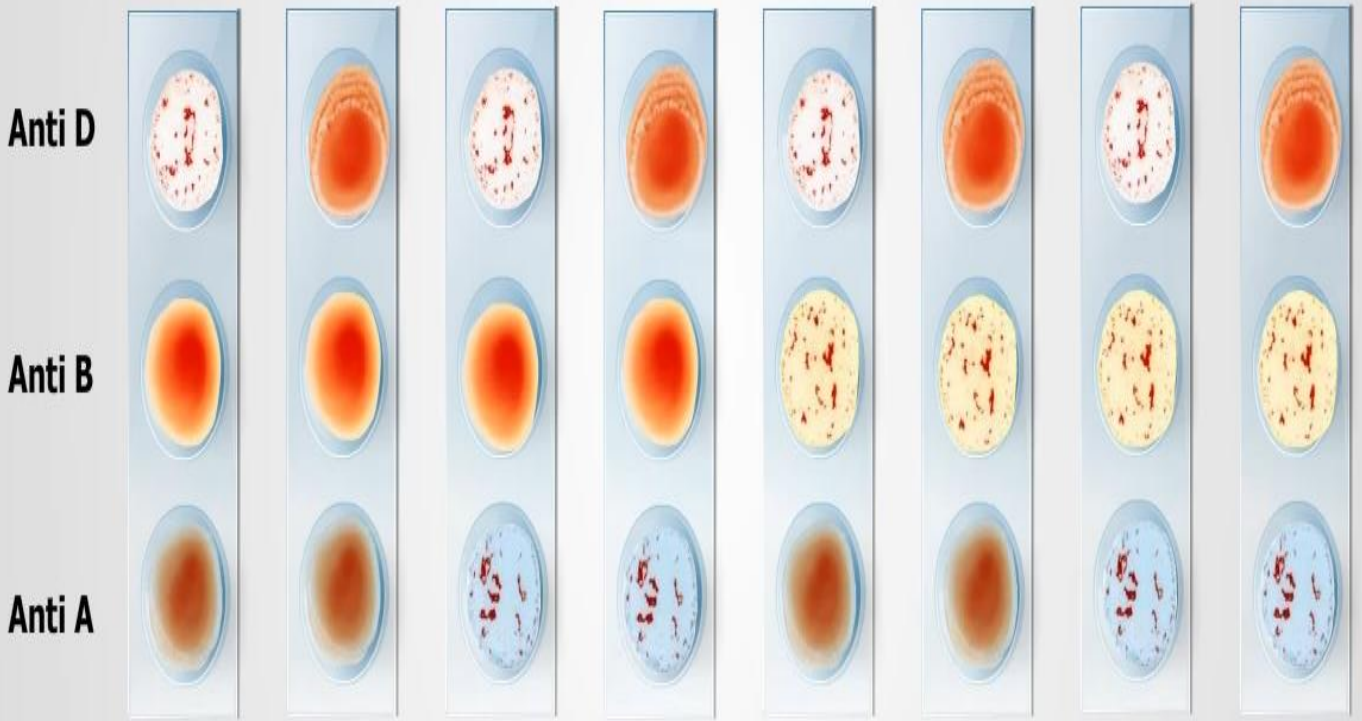





A+
B+
AB+
O −

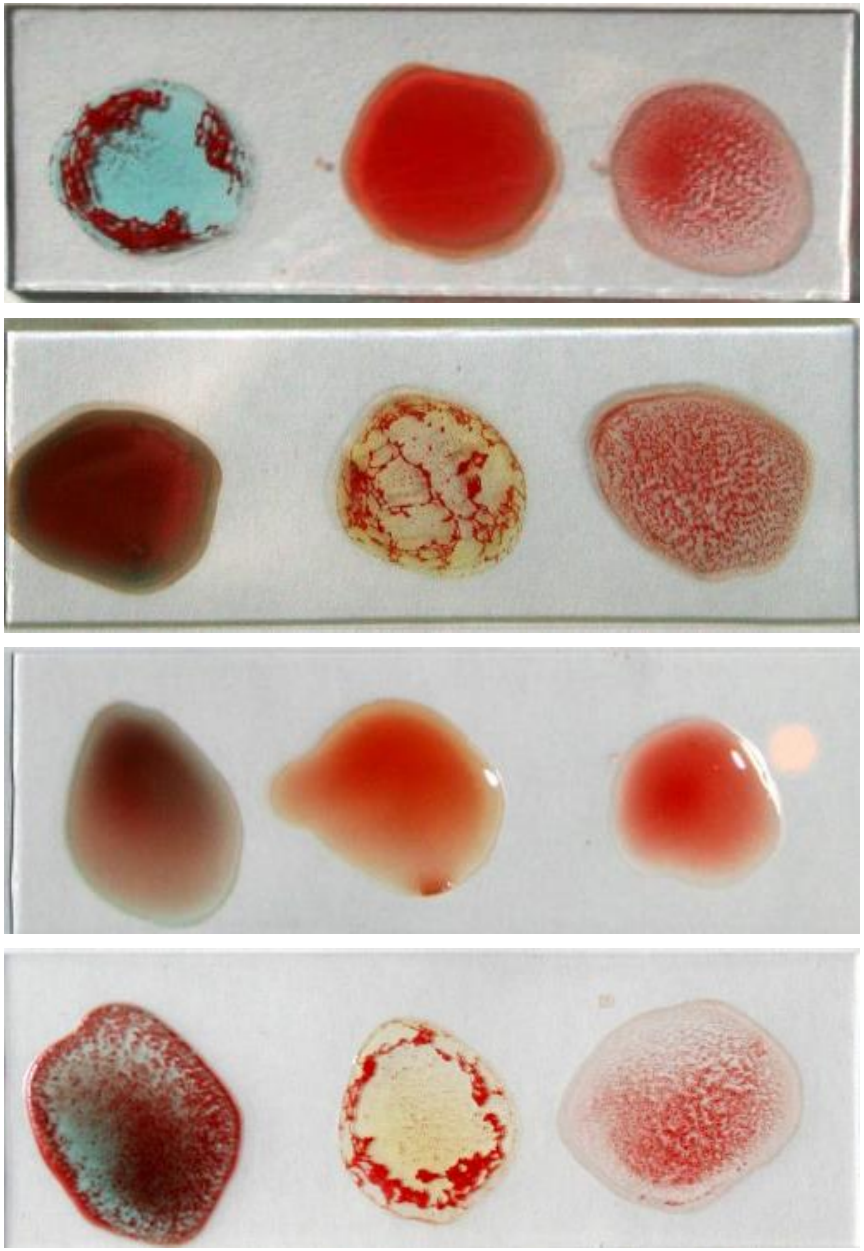


ABO Blood Groups





Anti A	Anti B	Anti D



Blood groups testing checklist

	Points of Evaluation	*Weight of each step (in marks)*	Marks of the student
1	The student put 3 drops of antibody (A), antibody (B) and antibody (C) on one side of the slide.	0.25	
2	The student put the slide on a white piece of paper and labeled the name of each antibody on it.	0.5	
3	The student put 3 drops of blood on the slide.	0.5	
4	The student mixed the blood and antibodies using a clean stick (stick for each mix).	0.5	
5	The student tilted the slide after 2 minutes to look for agglutination.	0.5	
6	The student determined the type of the blood group according to ABO and RH systems.	0.5	
7	The student discarded the used tools safely.	0.25	
Total marks		3	



Thank you